

ALI SHAHIDI

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SUMMARY

Software Developer with combined experience across Unity/XR systems, C#/.NET backend engineering, and AI pipeline development. Built production-style ASP.NET Core APIs with JWT security, Dapper/Entity Framework data access, and role-based workflows, while also delivering multiplayer VR/AR applications, multimodal inference systems, and RAG-enabled assistants. Designed agent-style LLM workflows (including ChatGPT Custom GPT-style setups with tool/API actions) orchestrated against .NET backends for simulation and training systems. Strong in real-time architecture, performance optimization, and translating research-driven prototypes into maintainable software for cross-functional teams.

PROFESSIONAL EXPERIENCE

Software Developer *September 2024 - November 2025*
ETHEREAL Research Group *Calgary, AB*

- Designed and developed an accessible VR training environment for non-speaking autistic users, enhancing interaction flow and ethical UX.
- Engineered backend systems to synchronize GPT-driven dialogue, user events, and multimodal sensory feedback in real time.
- Built GPT-agent prompting workflows that translated multimodal signals into clinician-oriented summaries, with privacy-minded handling before external LLM calls.
- Implemented AI-based behavioral analysis, boosting video processing speed by 90% and accuracy by 83%.
- Optimized performance and cut third-party API costs by 98% through adaptive frame rate control and efficient body-language delta analysis.

Software Developer *September 2024 - March 2025*
ELIXR & Illumia *Calgary, AB*

- Delivered VR/AR solutions across 5 projects, integrating Unity, C#, animation, AI, real-time audio, and WebXR to create immersive, interactive simulators.
- Developed cockpit and ATC training simulators with agent-style LLM orchestration, wiring simulation events to a .NET backend through API tool actions (Custom GPT-style configuration and instructions).
- Built Retrieval-Augmented Generation (RAG) pipelines using Llama 3 and Ollama, enabling context-aware querying of domain-specific documents.
- Developed real-time audio streaming and AI-driven animation systems, automating workflows and reflecting user emotions accurately.
- Optimized Unity performance with profiling, GPU instancing, batching, occlusion culling, and WebSocket streaming; refactored codebases to scale efficiently.

Software Developer - Internship *February 2024 - August 2024*
Telus *Calgary, AB*

- Led development of a cross-platform multiplayer AR app for iOS and Quest, enabling real-time shared experiences for up to 50 users.
- Engineered NetCode and spatial anchor systems for accurate mapping, animation, and interactive AR workflows.
- Enhanced operational efficiency, reducing transportation costs by 40% through immersive remote instruction.

Graduate Research & Teaching Assistant - Software Engineer *January 2021 - April 2024*
University of Calgary *Calgary, AB*

- Supported instruction in software engineering, object-oriented programming, and scalable data analysis.
- Built software prototypes and data workflows for assistive technology research with interdisciplinary collaborators.

EDUCATION

University of Calgary

Master of Software Engineering (GPA: 4/4.00)

April 2024
Calgary, Canada

- Received full departmental graduate funding and international graduate tuition award.
- Nominated for the Best Graduate Project at the Distinguished Master's Thesis Award in STEM Disciplines.

University of Tehran

B.Sc. in Computer Engineering (GPA: 17.24/20.00)

July 2020
Tehran, Iran

- Received the best undergraduate project, and two Faculty of Engineering Fellowship Awards.

PROJECTS

Social Networking API | ASP.NET Core, MVC, REST APIs, MSSQL, Dapper, Entity Framework

- Built a full-stack backend using ASP.NET Core and MSSQL following MVC architecture, exposing RESTful APIs for user management with Swagger documentation.
- Implemented secure JWT authentication with password hashing/salting, token expiration, and role-based authorization for user and admin workflows.
- Designed a scalable data access layer using Dapper and Entity Framework, supporting complex relational queries and real-time data persistence.

ETHEREAL Research Group - Coffee Shop Simulation | Unity, C#, .NET, AI, API, VR, SQL

- Developed a high-fidelity VR training environment for neurodivergent users, integrating multimodal sensory feedback and ethical UX design.
- Engineered a .NET backend to manage real-time application state and handle asynchronous API-driven communication between Unity and external LLMs.
- Integrated GPT-driven dialogue and ElevenLabs voice synthesis to facilitate adaptive, low-latency NPC interactions.
- Shaped simulation-based customer-service training where LLM output drove character animation cues and rubric-style session feedback for trainees.
- Implemented a vision-AI pipeline to parse visual stimuli and procedurally generate decision-tree UIs for complex user workflows.

ETHEREAL Research Group - Multi Modal Behaviour Analysis | Python, AI, Azure, Unity, VR/AR

- Architected a cross-platform real-time inference system for Meta Quest 3 and mobile tablets, synchronizing multimodal inputs (Vision, Audio, Text).
- Engineered a Python-based serverless pipeline on Azure to offload GPU-intensive computation, enabling high-speed processing of large-scale data via custom chunking and streaming logic.
- Optimized data ingestion layers to handle concurrent multimodal streams, ensuring sub-second latency between headset input and AI-generated feedback.
- Built GPT-agent style orchestration over synchronized vision, audio, and text streams to support decision-oriented interpretation beyond plain text summarization.

ELIXR & Illumia - ATC | Unity, C#, Network, Optimization, VR, Python, RAG, API

- Engineered a Retrieval-Augmented Generation (RAG) pipeline utilizing Llama 3 and Ollama for real-time querying of domain-specific ATC manuals, with metadata-filtered retrieval, top-k selection, and re-ranking for higher precision under noisy transcripts.
- Developed a low-latency audio streaming and transcription pipeline in Python, achieving 86% accuracy in command extraction across diverse accents.
- Combined RAG-grounded reasoning with deterministic NLP parsing so noisy ATC transcripts translated into reliable automated system actions, reducing trainee manual workload during high-stress simulations.

Internal Operations Portal (ETHEREAL) | ASP.NET Core, C#, REST APIs, SQL, DTOs, ViewModels

- Built an internal-use .NET portal for project tracking and workflow visibility, using ASP.NET Core APIs and structured DTO/ViewModel mappings.

- Implemented role-aware endpoints and validation patterns to keep business logic centralized and maintainable across portal modules.
- Improved operational clarity by consolidating status, task, and review flows into a single backend-driven interface.

Community Sim | Python, APIs, JSON, Automation

- Developed and maintained Community Sim in Python, integrating API workflows and structured JSON data exchange.
- Automated repetitive setup and validation tasks to improve release consistency and reduce manual overhead.
- Documented troubleshooting and operational handoff guidance for team reuse and maintainability.

SKILLS

Languages: C#, Python, C, C++, JavaScript, SQL

AI & Machine Learning: RAG Pipelines, Llama 3, OpenAI API, Custom GPT-style agents, tool/API actions, Multimodal Inference, NLP, Azure AI

XR & Systems: Unity, Meta Quest SDK, WebXR, NetCode (Multiplayer), Spatial Anchors, AR Foundation

Backend & Cloud: ASP.NET Core, MVC, RESTful APIs, Azure, Docker, Micro-ORMs (Dapper)

Databases: MS SQL, MySQL, PostgreSQL

Tools & Workflow: Git, Linux, LaTeX, Scrum/Agile, Jira, Confluence